Ilankoon Pathiranage Pabasara Vikum Ilankoon

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Professional Summary

Electronic and Telecommunication Engineering undergraduate at General Sir John Kotelawala Defence University with a strong interest in **Embedded Systems, Artificial Intelligence, Machine Learning, Computer Vision, IoT and Wireless Communication**. Experienced in developing innovative engineering solutions through projects involving **AI-powered detection systems, embedded hardware integration, cloud-connected applications and long-range communication networks**. Proficient in Python, C++, Raspberry Pi, ESP32, OpenCV, YOLO and related technologies, with strong analytical, problem-solving and teamwork skills demonstrated through multidisciplinary engineering projects.

Education

General Sir John Kotelawala Defence University — **Feb. 2024 – Present**
BSc. (Hons) in Electronic and Telecommunication Engineering

University of Colombo School of Computing — **Sep. 2024 – Present**
BSc. in Information Technology (External)

St. Anne's College, Kurunegala — **Jan. 2013 – Feb. 2022**
GCE Advanced Level (Physical Science Stream)
GCE Ordinary Level

Relevant Coursework

- Signals and Systems
- Electronic Systems
- Communication Theory
- Control Systems
- Electromagnetics
- Electronic Circuits
- Communication Networks
- Digital System Design
- Embedded Systems
- Digital Signal Processing
- Power Electronics
- Image Processing and Vision

Technical Skills

Programming	Python, C, C++, Arduino, JavaScript
AI/ML & Vision	Machine Learning, Deep Learning, OpenCV, YOLO, CNN, TensorFlow, Keras, MobileNetV2, Transfer Learning, Image Processing, Object Detection, Computer Vision, CVAT, Data Annotation
Embedded Systems	ESP32, ESP32-S3, ESP32-C3-Mini, STM32, Raspberry Pi, Arduino, Sensor Integration, Motor Control, Motor Drivers, PCB Design, Circuit Debugging, Power Systems
Robotics & Autonomy	Maze Solving Robots, Autonomous Navigation, Path Planning, Sensor Fusion, Obstacle Detection, PID Control, Real-Time Embedded Systems, PCB Designing, Robot Control Systems
Cloud & Software	Firebase, FastAPI, REST APIs, Flutter, React.js, Tailwind CSS, Git, GitHub, Mobile App Development, Web Application Development
Communication	LoRa, GSM, Wi-Fi, Bluetooth, UART, SPI, I2C, IoT Systems
Design Tools	EasyEDA, Proteus, Solidworks, LTSpice, MATLAB, VS Code, Arduino IDE, Microchip Studio, Tindercad

Research Experience

Elevision: A Real-Time AI-Powered Elephant Detection and Alert System for Railway Safety in Sri Lanka

- Researching an AI-based railway safety system for real-time elephant detection and early warning.
- Developing YOLOv8-based computer vision models for elephant detection in railway environments.
- Implementing edge AI deployment on Raspberry Pi using ONNX for real-time inference.
- Integrating Firebase, IoT communication, and Flutter applications for remote monitoring and alerts.
- Preparing a preprint manuscript on system design, implementation, and performance evaluation.

Keywords: Elephant Detection, YOLOv8, Railway Safety, Computer Vision, Edge AI, Firebase, Raspberry Pi, Flutter

Publication Status: Preprint Manuscript (Under Preparation)

Projects

Elevision | AI-Based Elephant Detection and Early Warning System

- Designed and developed a real-time AI-based railway safety system to detect elephants on or near railway tracks using computer vision and deep learning.
- Implemented YOLO-based object detection with OpenCV for real-time video frame analysis and wildlife detection.
- Built a web application for live monitoring, alert visualization, and system status tracking.
- Developed a Flutter mobile application for real-time alerts and remote monitoring.
- Integrated Raspberry Pi edge computing, GSM communication, and Firebase cloud services for real-time alerting and storage.

Technologies: Python, OpenCV, YOLO, CNN, Raspberry Pi, Firebase, GSM, IoT, Flutter, React.js

LankaMesh | LoRa-Based Disaster Communication System

- Designed and developed a decentralized emergency communication system for disaster environments without internet or cellular networks.
- Implemented LoRa-based long-range peer-to-peer communication for SOS alerts, messages, and GPS data sharing.
- Designed embedded system architecture using ESP32-S3, LoRa RA-02 (SX1278), and GPS modules.
- Built a Flutter mobile application for emergency alert categorization and communication handling.
- Demonstrated scalable IoT-based disaster communication framework for resilient connectivity.

Technologies: ESP32-S3, LoRa RA-02, GPS, Flutter, IoT, Embedded Systems, Wireless Communication

AgroVision Tomato AI | Intelligent Tomato Leaf Disease Detection System

- Developed a full-stack AI system for tomato leaf disease detection using deep learning and computer vision.
- Built a MobileNetV2 transfer learning model for multi-class plant disease classification.
- Developed web (React.js) and mobile (Flutter) applications for image upload and prediction results.
- Implemented FastAPI backend for authentication, prediction, and database management.
- Integrated Firebase cloud storage for image dataset handling and prediction history tracking.

Technologies: Python, TensorFlow, Keras, OpenCV, FastAPI, React.js, Flutter, Firebase, CNN, MobileNetV2

Digital Communication Simulator

- Developed a web-based simulation platform to model and analyze digital communication systems.
- Implemented modulation schemes including ASK, QPSK, 16-QAM, and OFDM for system analysis.
- Simulated AWGN channel effects to evaluate signal degradation and noise impact.
- Built BER vs SNR performance evaluation tools for communication system comparison.
- Designed interactive signal visualization tools using Plotly and Matplotlib.

Technologies: Python, Flask, NumPy, SciPy, Plotly, Matplotlib, HTML, CSS, JavaScript, DSP

Mars Robot Competition | Autonomous Robotics System

- Designed and developed an autonomous robotic system for competition-based navigation and object handling tasks.
- Implemented sensor fusion using IR sensors, ultrasonic sensors, gyroscope, and color sensors.
- Developed embedded control system using ESP32-S3 for real-time decision-making and motor control.
- Designed mechanical system using SolidWorks and 3D printing for structural fabrication.
- Implemented encoder-based motor control using TB6612FNG driver for precise movement control.

Technologies: ESP32-S3, C/C++, IR, Ultrasonic, MPU6050, TCS34725, SolidWorks, Embedded Systems

ROSCO'25 Competition Robot | Autonomous Line and Wall Following Robot

- Built an autonomous robot capable of line following, wall following, and ramp navigation tasks.
- Implemented sensor fusion using VL53L0X ToF sensors, IR array, and MPU6050 IMU.
- Developed embedded control system using ESP32-S3 and TB6612FNG motor driver.
- Designed chassis using laser-cut acrylic and custom 3D-printed mechanical parts.
- Integrated encoder-based feedback system for accurate motion control and stability.

Technologies: ESP32-S3, VL53L0X, IR Sensors, MPU6050, TB6612FNG, SolidWorks, Embedded Systems

Achievements

Champions	Target Sprint Shooting Competition
KDU Colours	Air Rifle - 2025
KDU Merit	Air Rifle - 2024

Certifications

MATLAB Fundamentals	2024, MATLAB Training Program; Numerical computing, matrix operations, data visualization, scripting, and engineering simulation techniques.
AI/ML Engineer	2026, SLIIT; Machine learning pipelines, supervised and unsupervised learning, neural networks, deep learning, TensorFlow, Keras, and model evaluation techniques.
Deep Learning Specialization	2024, DeepLearning.AI; Neural networks, CNNs, sequence models, optimization algorithms, hyperparameter tuning, and reinforcement learning fundamentals.

Leadership & Volunteering

Chairman: IEEE ComSoc KDU — June 2026 – Present

Finance Team Lead: GENESIZ'26 — May 2026 – Present

Finance Committee Member: ERIC KDU — Feb. 2026 – Present

Program Team Member: IEEE ComSoc KDU — June 2025 – June 2026

Design Committee Member: Mathematical Society KDU — Feb. 2025 – Feb. 2026

Professional Affiliation

IET Member: Institution of Engineering and Technology

References

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